

Appendix

A Implementation Details

A.1 Model Fine-Tuning

All Vision-Language-Action (VLA) policies in this work are derived from a pre-trained base model π_0 . The fine-tuning process is conducted as follows:

- **Parameter Update:** For both the hand-only and the arm-hand VLA policies, we perform full-parameter fine-tuning on π_0 , with the exception of the visual encoder, which remains frozen.
- **Task Instruction:** The specific language instruction provided to the model is tailored to the task:
 - Instruction for the **hand-only VLA** policy: pick up the object on the table and place it elsewhere.
 - Instruction for the **arm-hand VLA** policy: pick up the object on the table and place it in the box.

A.2 Hardware Platforms and Real-World Deployment

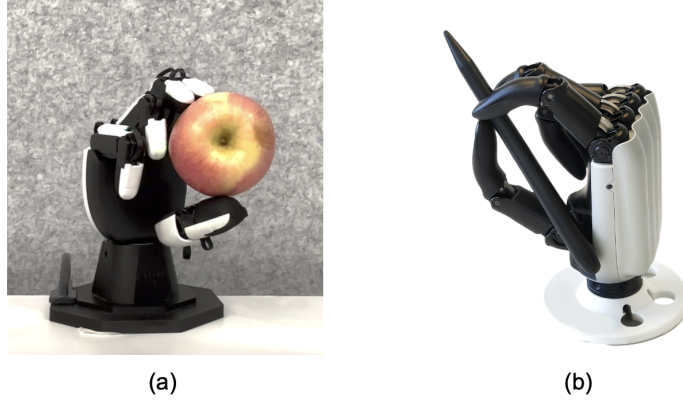


Figure 16 Two different robotic hands used in the study for validating the Shared Autonomy framework for training end-to-end VLA models: (a) XHAND1 hand [2], and (b) RY-H2 hand [1].

This research uses two types of dexterous robotic hands: (1) **Xhand**, a high-performance 12-DoF dexterous robotic hand, as the main hardware; and (2) **RY-H2**, a fast 11-DoF dexterous hand (6 active DoF and 5 under-actuated DoF) with a quick open-close cycle of 0.4s.

The **Xhand** provides a high-bandwidth motor control interface for learning-based control, joint-level state data, and high-resolution tactile feedback [2]. Its five fingertips are equipped with 270° encirclement of tactile array sensors, providing 120 channels in total for the 3-dimensional force data per fingertip, providing tactile perception of contact geometry and distributed forces. The motors offer multiple control modes, including position, force, and hybrid force-position control, running at a control frequency of 83 Hz over an EtherCAT. With a fingertip grip force of 15 N and a maximum grip force of 80 N, the hand can perform both precise and powerful grasps. Its design of back-drivable actuation and a lifetime of 1000000 grasp cycles, makes it suitable for the extensive trial-and-error required in data collection and testing trials of VLA policies.

The **RY-H2** hand is a five-finger under-actuated dexterous hand featured by high-speed grasping [1]. Totalling 11 joints, with 6 active and 5 passive degrees of freedom, it is actuated by high-power-density brushless DC motors. This design enables a rapid open-close cycle time of 0.4s and a high maximum grip force of 140N. With a lightweight of 0.6kg, it is suitable for dynamic tasks requiring both speed and power, and secondary algorithmic development for industrial and research applications.

During the evaluation phase on the physical robot system, the following configurations are applied:

- **Model Checkpoint:** All VLA models evaluated in the main experiments and the appendix are the checkpoints saved at 80,000 training steps.
- **Control Frequency:** The robot arm and hand are controlled by the policy at **30 Hz** control frequency.

A.3 Network Architecture Specifications

The Arm-Hand Feature Enhancement module extends the base architecture with dedicated components for limb-specific feature extraction. The arm encoder $\mathcal{E}_{\text{arm}}$ and the hand encoder $\mathcal{E}_{\text{hand}}$ are both implemented as a two-layer MLP. Each encoder takes the shared representation $z_t^{\text{share}} \in \mathbb{R}^{d_s}$ as input and produces limb-specific features of reduced dimensionality $z_t^{\text{arm}} \in \mathbb{R}^{d_s/2}$ and $z_t^{\text{hand}} \in \mathbb{R}^{d_s/2}$ through successive linear transformations separated by the Mish activation functions.

The auxiliary prediction heads $\mathcal{H}_{\text{arm}}$ and $\mathcal{H}_{\text{hand}}$ are implemented as single linear layers that map the limb-specific features to action predictions of a fixed max dimension, with selective supervision applied only to the indices corresponding to each limb’s actual degrees of freedom.

B Additional Results

B.1 Effectiveness of Shared Autonomy Data Collection

Table 4 Data collection efficiency and training/deploying expenditure of shared autonomy vs full teleoperation.

Methods	Main Collection	Corrective Collection	Fine-Tuning Time	Deploying Time
Shared Autonomy	110/hour/person	100/hour/person	4 hours (on 4 GPUs)	10–15 minutes (20 trials)
Full Teleoperation	90/hour/person	80/hour/person	-	-

Our Shared Autonomy framework demonstrates a clear advantage in data collection efficiency. As shown in Table 4, it allows a single operator to collect 110 trajectories per hour for the main dataset, compared to 90 with full teleoperation. This 25% increase in collection rate, sustained during corrective data collection (100 vs. 80 trajectories/hour), directly translates to faster policy improvement cycles and validates the framework’s effectiveness.

This high-quality data allows for efficient policy refinement: a fine-tuning run (20k steps on 4 GPUs) completes in 4 hours, and deploying the refined policy for 20 evaluation trials takes only 10–15 minutes. This end-to-end efficiency shows the feasibility of our approach for rapid policy training and iteration. For skilled operators, collecting more than 100 demos per hour per person is easily and readily achievable, and most domain-specific cases usually require around 50 demos only, which enables a development-to-deployment cycle of one day only.

B.2 Additional Results of End-to-End Adaptive Arm-Hand Grasping

This section presents additional qualitative results, and Fig. 17 to Fig. 19 demonstrate the robustness and generalization capability of our end-to-end VLA policy in additional grasping scenarios.



Figure 17 Grasping performance across different object orientations.

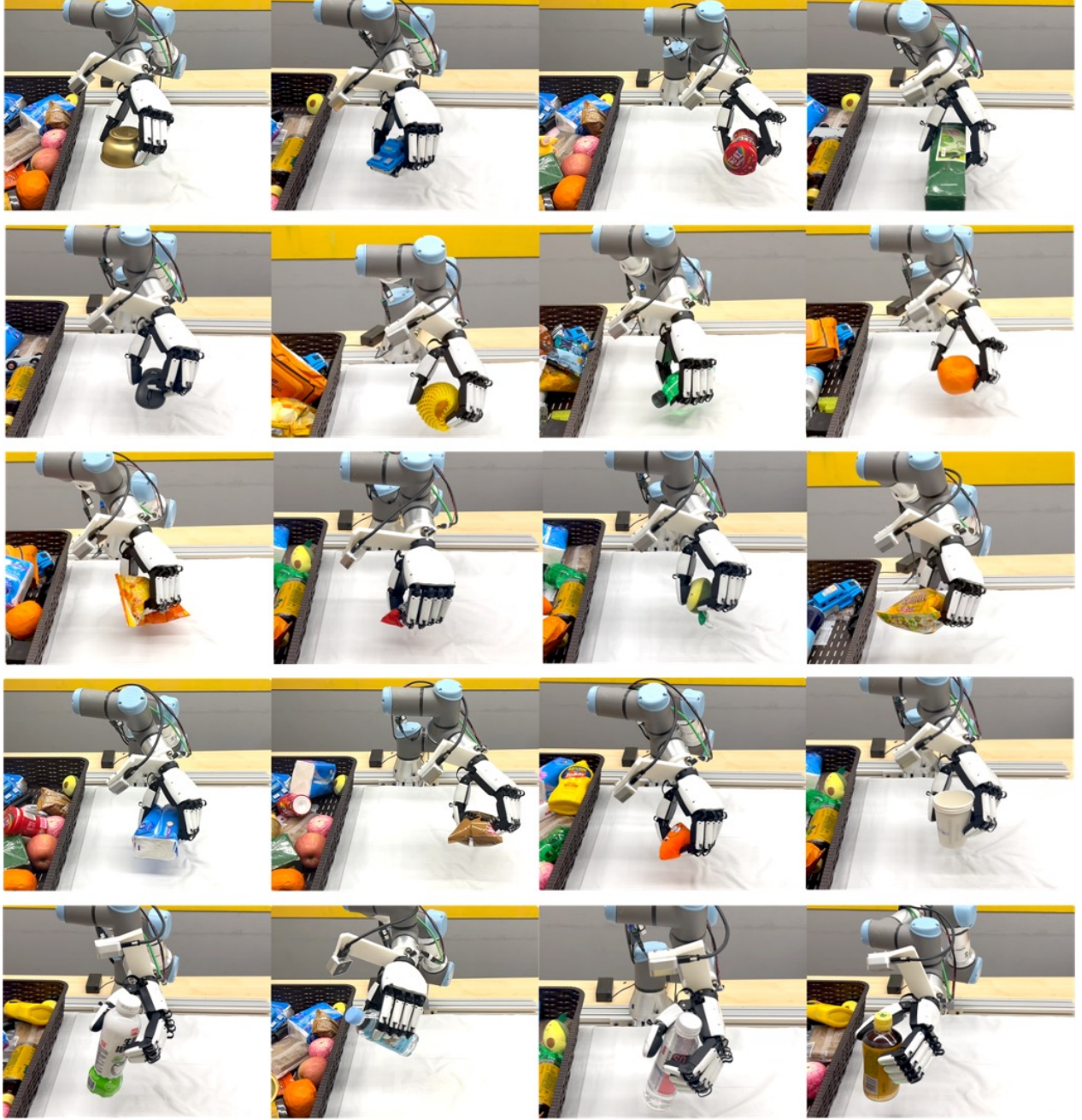


Figure 18 Grasping diverse objects with variations in size, color, and material properties.

B.2.1 Effectiveness of Tactile Sensing in π_{uni}

To evaluate the impact of tactile sensing on the full arm-hand policy, we conducted an additional experiment under the same conditions as our main $\mathbf{X}_{\text{hand}}$ evaluation (20×20 cm workspace, 10 objects, 10 trials each). The tactile representation and integration method followed the same approach as in the hand-only DexGrasp-VLA policy. Despite the tactile sensing provides significant benefits to the hand-only policy π_{hand} , we found that incorporating these same tactile features into the unified arm-hand policy π_{uni} does not consistently improve performance. In other words, tactile sensing is more effective when used in combination with local visual sensing for the hand-only DexGrasp VLA policy, but the direct incorporation of tactile sensing in the unified arm-hand policy π_{uni} does *not* yield positive results, at least from this initial study. Specifically, the enhanced policy with tactile input ($\pi_{\text{uni-enhance-tac}}$) achieved a success rate of 82%, compared to 95% for the



Figure 19 Grasping robustness tested at various spatial locations within the workspace.

visual-proprioceptive only policy ($\pi_{\text{uni-enhance}}$) trained by datasets collected by shared autonomy.

This performance degradation is likely due to the different functional roles for controlling the arm and the hand respectively. Unfortunately, due to the lifetime and closure of this project, no continued efforts could be pursued for further investigation. To the best of our knowledge, our current understanding is shared as follows. The arm primarily executes reaching motions that rely more on visual and proprioceptive feedback for spatial guidance, while tactile signals are most relevant for fine-grained grasping and in-hand manipulation. Uniformly incorporating tactile input throughout the entire arm-hand trajectory may introduce irrelevant information like “noises” during arm movement phases, particularly from incidental environmental contacts (e.g., table collisions or unintended fingertip brushing) that occur during reaching. These transient and often misleading tactile signals appear to interfere with the policy’s ability to maintain robust arm-centric coordination.

The above results suggest that future work should explore more structured tactile integration strategies rather than uniform feature fusion throughout the entire motion. Promising directions include selective sensor gating mechanisms that activate tactile processing only during grasping phases, or attention-based architectures that learn to dynamically weight tactile input based on the current task phase. Such approaches could preserve the benefits of tactile sensing for manipulation, while avoiding the performance degradation observed during arm movements.

B.2.2 Additional Results of Corrective Control for Refining a VLA Policy

This appendix extends the experimental validation of our corrective framework beyond the pick-and-place tasks presented in the main text. Together with the shared autonomy approach used for grasping tasks in the main experiments, the additional studies here, which employed teleoperation for long-horizon tasks and

motion planning for industrial assembly, provide comprehensive evidence for the generality of our corrective mechanism across diverse tasks and data collection methodologies.

B.3 Long-Horizon Manipulation with Robotic Gripper

B.3.1 Long-Horizon Tasks Learned from Teleoperation

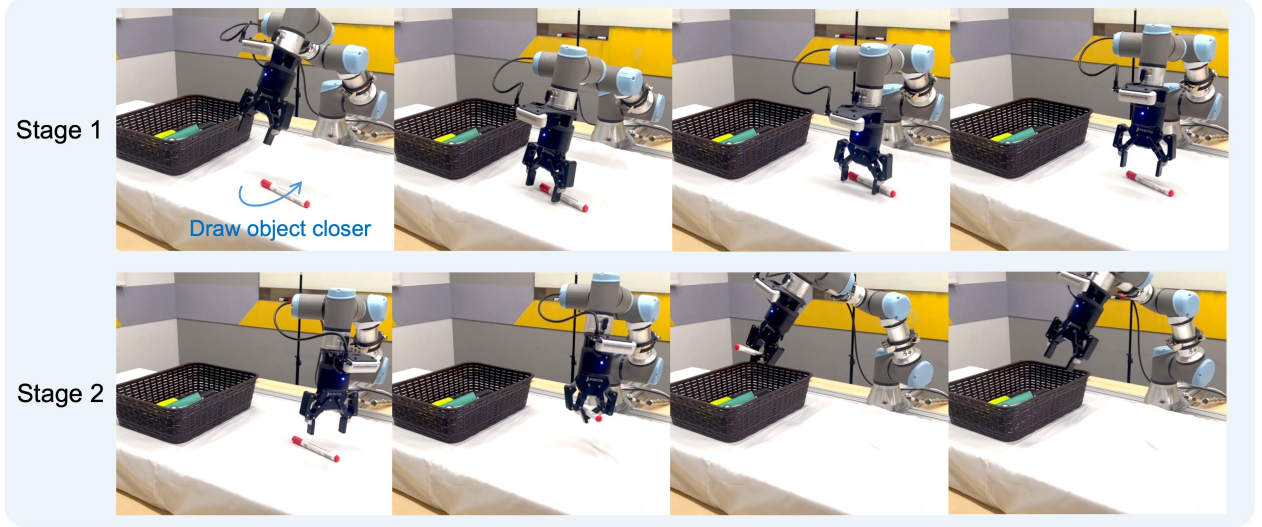


Figure 20 Task 1. Sequential object manipulation via multi-stage physical interactions with a distant object. The task involves a two-stage strategy to manipulate a pen from a distant, non-graspable location (at the arm’s far reach near singularity if performing the usual grasp) to a nearby location suitable for grasping. Stage 1: the robot sweeps, reorients, and draws the object closer to its base. Stage 2: the robot executes a final grasp and placement.

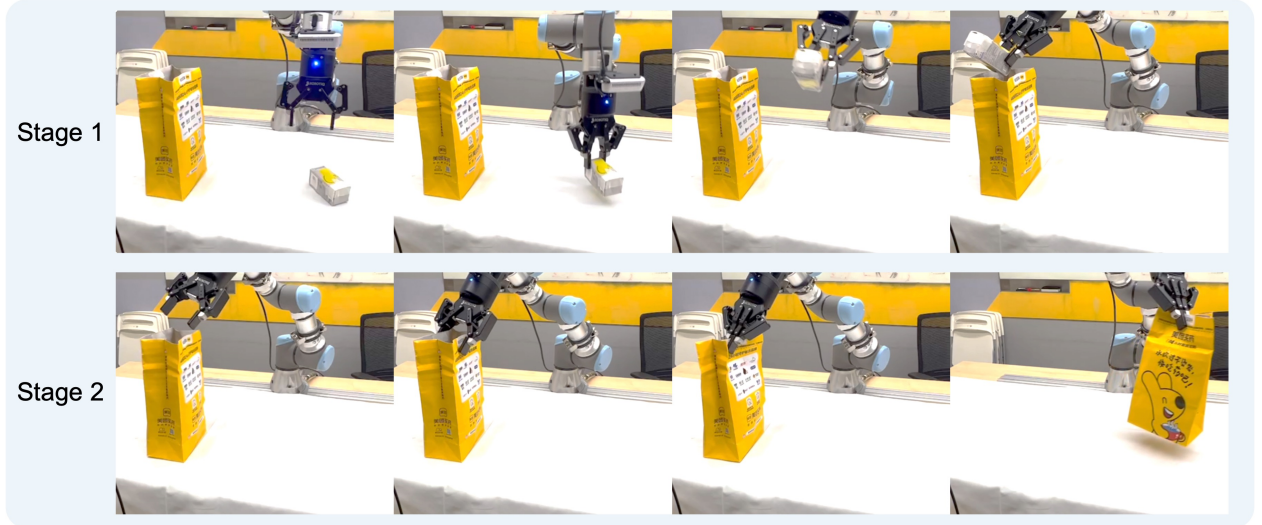


Figure 21 Task 2. Sequential multi-stage packing showing key actions. Stage 1: placing the medication box into the packaging bag. Stage 2: execution of closing the bag and transporting the closed package, showing the capability of operating both rigid and deformable objects.

Building upon the corrective framework validated in the main text’s using dexterous hands, we further designed three sequential long-horizon tasks using parallel jaw grippers to evaluate the framework’s efficacy under different hardware and task settings. These three tasks include:

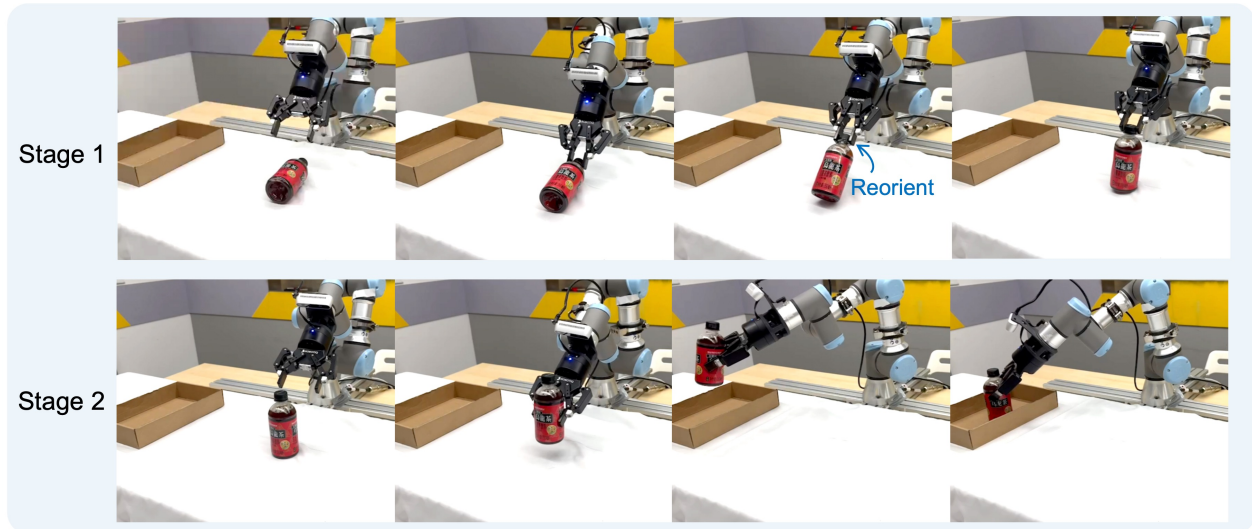


Figure 22 Task 3. Underactuated bottle uprighting by exploiting low torsional friction. Stage 1: grasping the cap and lifting the bottle to induce passive uprighting via low torsional friction around the cap. Stage 2: subsequent pick-and-place from a re-configured graspable pose to the target box.

Task 1. Pen Relocation and Placement (Fig. 20): This task requires the robot to first sweep a distant slender pen and bring it to a closer range, then to reorient the gripper to an appropriate angle, grasp and place the pen. Common failure modes include inadequate sweeping force and incorrect gripper orientation during the pre-grasp phase. In this scenario, the robot needs to first re-configure the object and change it from a non-graspable state into a graspable pose, which requires the design of RL policies previously with handcrafted reward design [45] and now can be learned through a VLA-based imitation by providing demonstrations for learning such behaviors.

Task 2. Pill Box Packing (Fig. 21): In this multi-stage task, the robot must sequentially grasp a small medication box, put it into a packaging bag, securely close the bag, and finally transport the entire package to a designated location. Failures typically occur during the delicate bag-closing phase and when handling the combined object during transport.

Task 3. Bottle Uprighting and Placement (Fig. 22): This challenging task begins with re-configuring the state of the object – grasping the cap of a horizontally positioned water bottle, followed by carefully lifting and uprighting the bottle into a vertical orientation – and then completes with a standard pick-and-place operation. The uprighting process presents particular difficulties in maintaining a stable grip, taking advantage of the low torsional friction for passive rotation of the bottle during the execution of the lifting trajectory.

Following our established framework, we first trained a base policy π_{base} on initial demonstrations. Then, we collected corrective trajectories via human teleoperation for those occurred failures across these diverse task scenarios. After fine-tuning, the resulting policy π_{corr} achieved significantly higher success rates (see Table 5), demonstrating that the corrective mechanism (as previously shown effective with shared autonomy) also works well for the hardware system using teleoperation in complex multi-step tasks with parallel jaw grippers.

Table 5 Success rates of long-horizon tasks via corrective teleoperation.

Tasks	Task 1 (Fig. 20)	Task 2 (Fig. 21)	Task 3 (Fig. 22)
Success Rates	65%	90%	70%

B.3.2 Learning Industrial Assembly Task - Data Collection through Motion Planning

Complementing the human-in-the-loop approaches (such as the shared autonomy in the main text and the teleoperation in long-horizon tasks in the Appendix), we investigated how *automated motion planning*

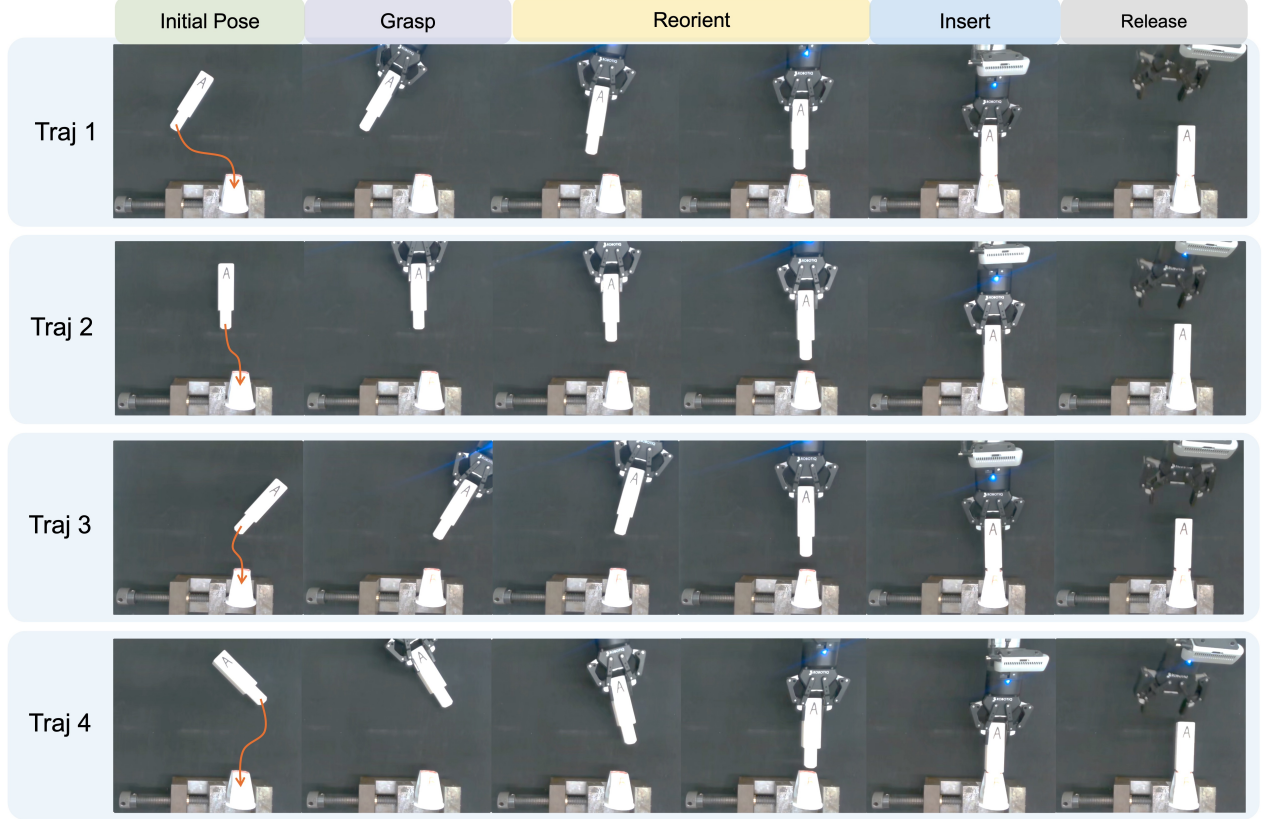


Figure 23 Peg-in-hole assembly task showing multiple stages (grasp, reorient, insert, release) under four different initial configurations.

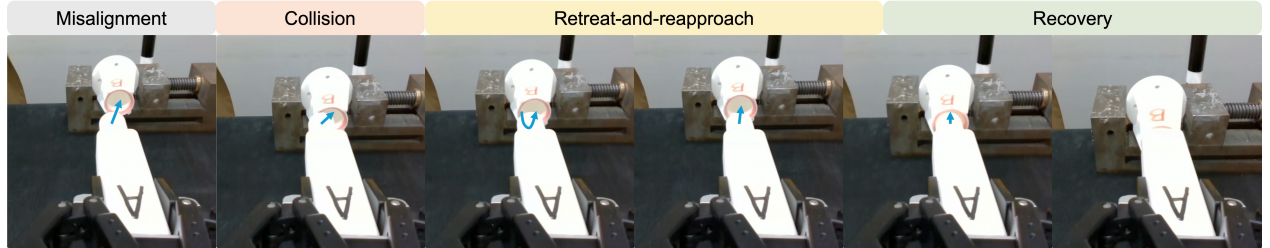


Figure 24 Time-elapsing frames of error-recovery during peg-in-hole assembly, showing stages of misalignment, collision, retreat-and-reapproach, and failure recovery with successful insertion.

could generate effective failure-correction and serve as a useful source of data. We applied our framework to a peg-in-hole assembly task – a canonical industrial task that requires spatial accuracy beyond typical pick-and-place. Following the same experimental protocol, we began with 100 initial demonstrations to train π_{base} using motion planning, where the representative cases from data collection are shown in Fig. 23.

When the policy failed on precision-critical cases, we employed **automated motion planning** to generate corrective trajectories (Fig. 24), instead of human intervention. The motion planner produced recovery behaviors, including fine adjustments and retreat-and-reapproach motions. Reported as in Table 6, after incorporating 20 additional recovery trajectories and fine-tuning, π_{corr} achieved a 90% success rate: **20% absolute improvement** over π_{base} . This result confirms that the corrective mechanism works effectively even with fully automated correction and its generated data sources, extending its applicability beyond human-guided interventions.

Table 6 Success rates of the task of peg-in-hole.

Methods	π_{base}	π_{corr}
Peg-in-Hole	70%	90%

B.4 Summary and Discussion

The collective evidence from all experiments – spanning pick-and-place tasks with shared autonomy (main text), long-horizon tasks with teleoperation, and industrial assembly with motion planning – consistently demonstrates the effectiveness of our corrective framework across task types and data collection approaches. Each study followed the same fundamental methodology: initial policy training, failure identification, corrective data collection, and policy re-training/refinement, yet employed different failure-correction methods and data sources tailored to task requirements.

This progression from human-guided corrections (e.g., shared autonomy, teleoperation) to fully automated solutions (e.g., motion planning) highlights the framework’s core versatility, which can be tailored to different functional components for a wide range of tasks. The shared autonomy approach balances human expertise with automated assistance for efficient grasping corrections; teleoperation provides full human control for complex multi-step tasks; while motion planning offers a fully automated solution for structured industrial environments.

Most importantly, all three approaches yield significant performance improvements, confirming that the corrective mechanism itself is the key driver of policy refinement and enhancement, while any specific data collection method merely serves as a means to this end. This inherent versatility makes our framework applicable across a broad spectrum of robotic learning scenarios, from human-centric environments to structured industrial settings. By providing an effective means for continuous improvement, our work paves the way for more capable and robust general-purpose VLA policies, ultimately expanding the reach of advanced autonomous systems in the real world.